

TEST & MEASUREMENT

Diverse application areas

Digital power analysers are used to measure voltage, current, wattage, apparent power, frequency and power factor of any electrical and electronic appliance, compact fluorescent lamps, light emitting diode bulbs, transformers, fans, air-conditioners, motors, auto parts and so on. High-accuracy, true-RMS, digital-readout and light-duty meters are used in general industrial maintenance, production, quality control and laboratories.

Anuj Rohilla, deputy general manager - products, SPI Engineers Pvt Ltd, points out "production and diagnostics in load industry as well as actuators and contractors" as another area of application of power analysers. He adds, "These serve as very efficient tools for increasing productivity and life expectancy of load devices and machine analysis with load impact."

With increasing focus on electric vehicles, energy consumption efficiency of quick chargers is an important measurement and evaluation criteria that is being sorted out by power analysers.

3390 from Hioki displays voltage, current, active power, apparent power, power factor, frequency, loss, efficiency, distortion factor and voltage imbalance rate (when measuring three voltages and three currents), all at the same time.

Energy is one of the most precious resources. Design engineers are under constant pressure to increase efficiency and reduce power consumption down to the last milliwatt. Challenging programmes like light-emitting diode and high-intensity discharge lighting, solar panel energy output, efficiency testing on inverters and pulse-width modulation motor drive systems on electric vehicles, all require fast, precise and reliable power measurement.

GPA62 from PCE allows energy measurement in symmetrically-loaded three-phase systems. Power analysers PCE-PA 8000, PCE-360 and PCE-830 support professional power and energy measurements in single- and three-phase systems.

Benchtop versus handheld

With the shift to handheld equipment, benchtop equipment were supposed to take a serious hit. However, "Benchtop instrument will always remain the choice for lab (R&D) and production testing," says Tripathi. Due to portability, handheld equipment can be taken out in the field and real-time tests can be performed more efficiently.

With the added over-the-air functionality and use of computers for processing data, the ease has been significantly increased. May be, sometime in the future, the only thing you would need to take out in the field would be probes. **EFY**

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